

Applied NLP

DATA 641

Lecture 14 — Latent Dirichlet Allocation for topic modeling

*Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems 1st Edition, Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana, O' Reilly and Natural Language Processing in Action, Understanding, analyzing, and generating text with Python, Hobson Lane, Cole Howard, Hannes Hapke, First edition, MANNING

Another Topic Model to Explore

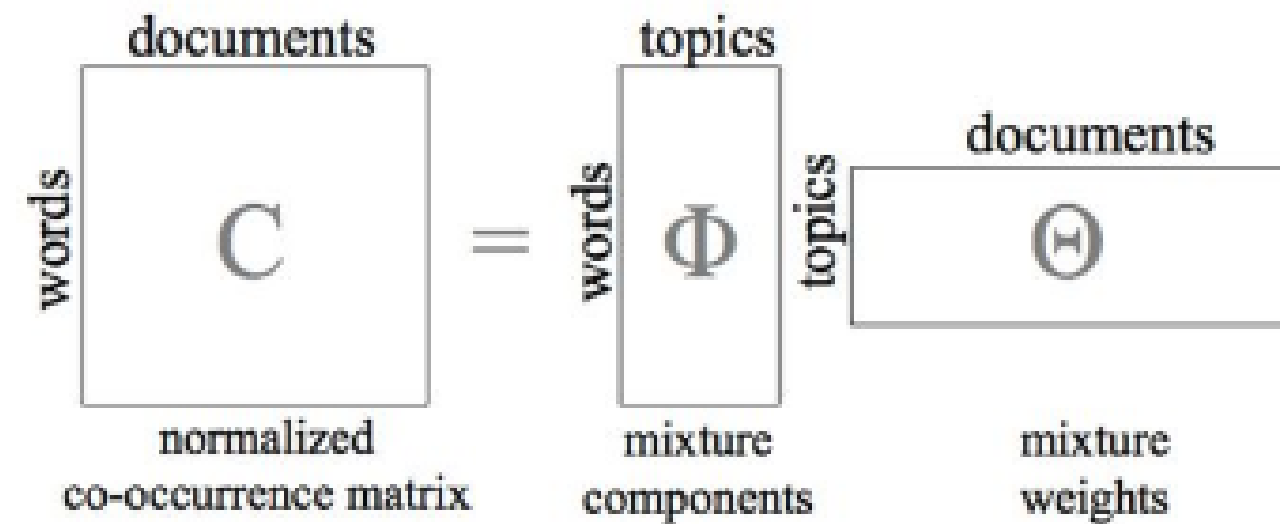
We discussed matrix factorization of the document-term matrix as a means of topic modeling

- Latent Semantic Allocation via Singular Value Decomposition (SVD)
- Non-Negative Factorization (NMF)

We did not cover Latent Dirichlet Allocation (LDA): a generative statistical model

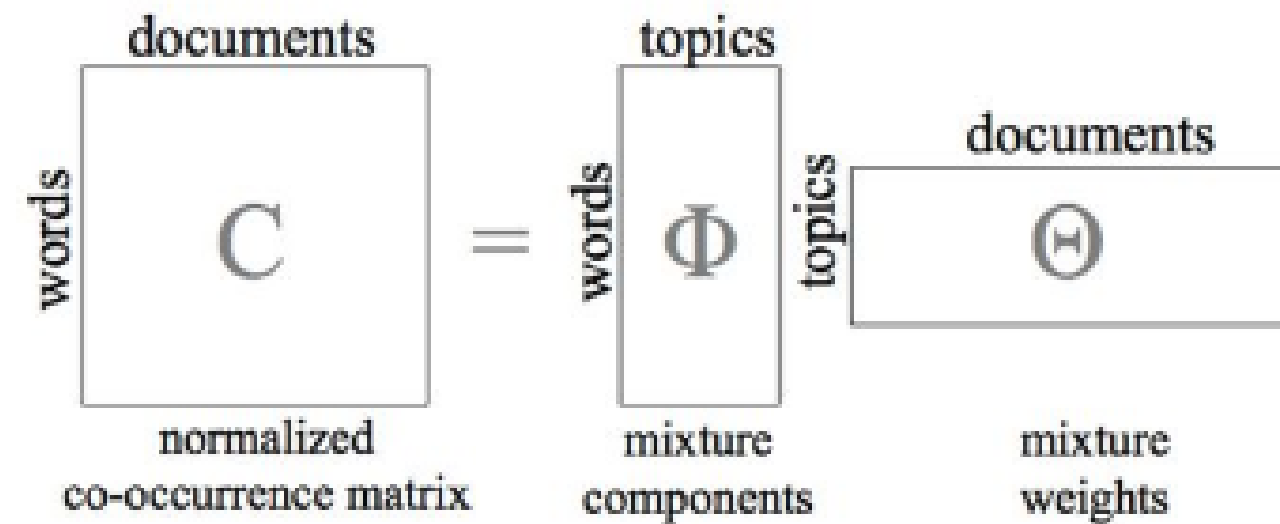
- **Documents as Random Mixtures over Latent Topics:** In topic modeling, documents are viewed as blends of hidden topics. Each document is not limited to a single topic but is a combination of various underlying themes. This approach allows us to uncover the fundamental subjects discussed in documents, providing a richer understanding of content.
- **Sparse Dirichlet Priors:** Sparse Dirichlet priors are used in models like Latent Dirichlet Allocation (LDA) to encourage sparsity in topic distributions within documents. They assume that most documents are influenced by only a few dominant topics, and most words are associated with only a subset of topics. This sparsity assumption helps reveal the most relevant topics in a document, enhancing the interpretability of topic modeling results.

Latent Semantic Analysis via LDA



- The Φ matrix represents the term-topic distribution. Each column of Φ corresponds to a topic, and each row represents a unique term or word in the vocabulary.
- The elements (ϕ_{ij}) represent the probability of observing term i in the context of topic j . In other words, it indicates how likely a particular term is to be associated with a specific topic.
- Mathematically, ϕ_{ij} represents $P(w_i|z_j)$, where w_i is a word, and z_j is a topic.

Latent Semantic Analysis via LDA



- The Θ represents the topic-document distribution. Each column of Θ corresponds to a document, and each row represents a topic in the topic space.
- The elements of Θ (θ_{ij}) represent the probability of a topic i being present in document j . In other words, it indicates how likely a document is to be associated with a specific topic.
- Mathematically, θ_{ij} represents $P(z_i|d_j)$, where z_i is a topic, and d_j is a document.

Underlying mechanism of LDA

Initialization:

- LDA starts with a collection of documents, where each document is represented as a bag of words (i.e., word frequency counts).
- It also requires the user to specify the number of topics, k , that the model should discover.

Initialization of Parameters:

- For each document, LDA initializes two matrices: the document-topic matrix Θ and the topic-word matrix Φ .
- Θ is initialized with random values, and Φ is also initialized with random values. These matrices represent the initial assumptions about the distribution of topics in documents and the distribution of words in topics.

Iterative Inference:

- LDA uses an iterative process to estimate the values of Θ and Φ that best explain the observed documents.
- During each iteration, LDA iterates over each word in each document and updates the topic assignments for that word based on the current estimates of Θ and Φ .
- These iterations continue until convergence, at which point the final estimates of Θ and Φ are obtained.

Output:

After convergence, LDA provides two main outputs:

- For each word in each document, LDA assigns a topic based on the final estimates of Θ and Φ . This gives us the topic composition of each document.
- LDA provides the topic-word distribution, which represents the probability of each word belonging to each topic. This distribution allows us to understand the most probable words associated with each topic.

Advantages of NMF for Topic Modeling

- **Interpretability:** LDA provides interpretable topics as its primary output. These topics are collections of words that represent coherent themes, making it easy to understand the main subjects within a corpus of documents.
- **Flexibility:** LDA is a flexible model that can be applied to various types of text data, including documents, articles, social media posts, and more. It adapts well to different domains and languages.
- **Unsupervised Learning:** LDA is an unsupervised learning method, meaning it does not require labeled data for training. This makes it suitable for tasks where labeled data is scarce or unavailable.
- **Topic Evolution Analysis:** LDA can be used to analyze how topics change or evolve over time by applying the model to different time periods. This is valuable for tracking trends and shifts in content.
- **Feature Reduction:** LDA reduces the dimensionality of the data by representing documents as mixtures of topics. This can be helpful for downstream tasks such as text classification and clustering.

Disadvantages of SVD for Topic Modeling

- **Number of Topics:** Choosing the right number of topics is challenging and can result in less meaningful or an excessive number of topics.
- **Fixed Topics:** LDA assumes static topics throughout the corpus, ignoring topic evolution or change.
- **Bag-of-Words Limitation:** LDA relies on bag-of-words representation, disregarding word order and document structure, limiting its understanding of context and semantics.
- **Sparsity:** In large datasets, the topic-word matrix can be extremely sparse, making it harder to extract meaningful insights.
- **Single-Level Model:** LDA is a flat model and does not naturally handle hierarchical topic structures found in many documents.
- **Preprocessing Sensitivity:** LDA's performance is sensitive to preprocessing quality, including tokenization, stop-word removal, and stemming.
- **Hyperparameter Tuning:** Effective use of LDA requires careful tuning of hyperparameters, such as the number of topics and priors, which can impact result quality.